

Vertical AI Agents

— From coding to tax

AI matured first in software. The same playbook is coming to tax — here's the map, what we're building at Meta, and what we got wrong along the way.

Less a roadmap than a year of learning.

J Gu · PM, Tax & Finance @ Meta

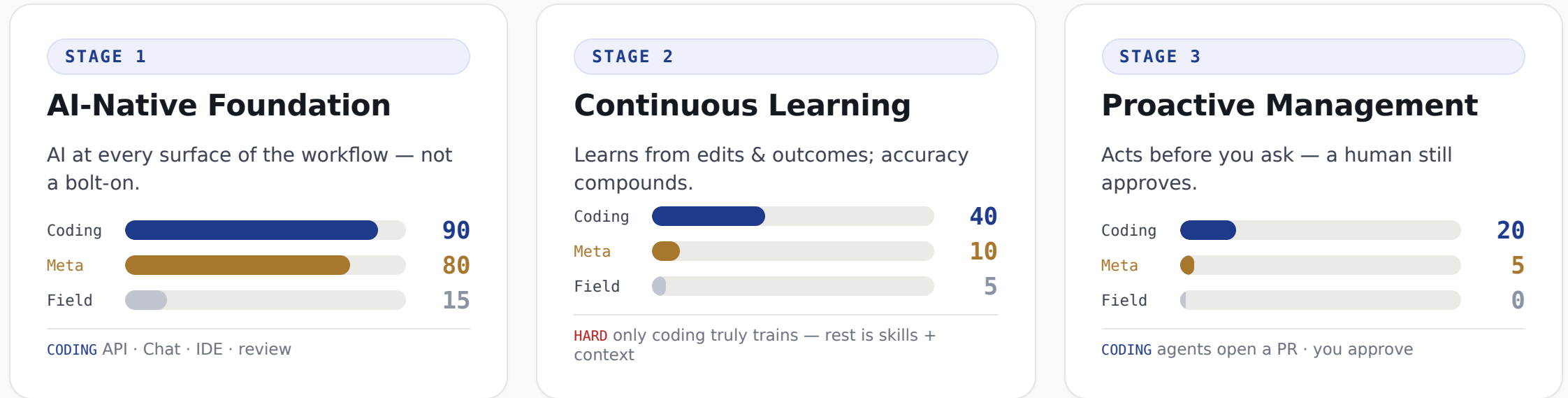
The arc — four quick acts, then your pushback

- | | | |
|----|--------------------|--|
| 01 | The pattern | Coding is the leading indicator — AI evolves in three stages |
| 02 | Our bet | One AI foundation across the whole tax lifecycle |
| 03 | The reality | 40+ skills in a year — and what humbled us |
| 04 | The how | The infrastructure an AI agent actually needs |

And throughout — **argue with me.** Most of this is still a work in progress.

The same three stages — coding leads, tax follows

AI's first act was chat; the next is **coding** — real work, not just talk. Coding is our leading indicator. Here's coding vs **Meta's tax build** vs the broader tax field (0-100).



We think we're ~80% there on the foundation — probably wrong on the exact number. Stage 2 is the frontier for everyone. Meta = projected year-end ·

Field = most tax teams today.

Coding figures it out first — then it diffuses

THE FRONTIER

Coding

Most investment, most talent — the playbook gets invented here first.

DIFFUSES



DOMAIN BY DOMAIN



Tax

startups now starting to build here



Legal

contracts, research, review



Healthcare

notes, coding, claims



Finance & more

each its own vertical

Each domain is its own journey. Tax is one — and the first **tax-AI startups are already appearing**. The opportunity is real; the foundation is the work.

Tax professionals are about to live what engineers already are

Coding agents didn't replace software engineers — they **amplified** them. AI will do the same for tax.

SOFTWARE ENGINEER • TODAY

+ coding agent

- Agents write the boilerplate & repetitive code
- The engineer directs, reviews & decides
- Freed for **architecture & judgment** — ships far more

SAME
STORY

TAX PROFESSIONAL • NEXT

+ tax AI

- AI does the assembly across systems & the repetitive work we outsource today
- The professional directs, reviews & signs off
- Freed for **planning, strategy & advising** — covers far more

No one's replaced. The work moves **up** — to judgment, strategy, and impact.

So we start at Stage 1 — across the whole tax lifecycle

We're building **one** AI-native foundation — and every stage of the tax lifecycle draws from it.

Tax Planning

Scenario modeling & structuring, grounded in our positions and the law.

Compliance & Reporting

High-volume, rules-based work on real-time data — the natural first mover.

Tax Audit

Controversy & defense on the same lineage, sources, and document trail.

1

One AI-Native Foundation shared data · grounding · governance — under all three

Foundation first. Build it once, build it well.

From a standing start to 40+ skills in a year

Q2 '25

foundation journey began

12 mo

in, and compounding

40+

skills shipped

3

lifecycle areas live

Planning

Skills for **new product initiatives** — model the tax impact of a launch before it ships.

Compliance & Reporting

10+ countries in testing now — **live by year-end** with very high coverage.

Audit — an end-to-end skill

1

One-click upload

>

2

Extract audit-critical info

>

3

Auto-search every DB & internal system

>

4

Draft the audit response

>

5

Reference historical audits

Real skills, in production — and each one makes the foundation stronger.

Three things that humbled us

1 • “Just ask the AI to do it”

WE ASSUMED

Hand it the audit — it writes the response.

WE LEARNED

AI is an intern. You teach it where the data lives, the reply format, your risk criteria — and document all of it. It's a long learning curve.

2 • “What's an Eval?” — nobody knew, 9 months ago

WE ASSUMED

Past audit responses are our answer key.

WE LEARNED

Last year's answer isn't valid this year — **even for the same question.** Evals have to be built on **today's fact patterns.**

3 • Context is not data

WE ASSUMED

Drop it all in Google Drive — even a spreadsheet — and call it knowledge.

WE LEARNED

Drive is fine for **context** (unstructured knowledge). But **data belongs in a database**, not a doc — that's why the foundation & graph matter.

AI isn't magic. It's an intern you have to train — and **we're still learning how.**

Humans in the loop — the engine behind continuous learning



No true continuous learning — yet

- Today we turn the wheel **by hand** — people correct, the AI improves via **skills + context**.
- Only the **coding** vertical actually **trains the model** in the loop.
- For tax — and every other vertical — real continuous learning is **unsolved, and hard to scale**.

Human-in-the-loop is how we improve now — and the **data engine** for continuous learning when it arrives.

Treat AI like a brilliant intern

Fast, capable, eager — and on day one it knows **nothing about your world**. You don't just hand it the work. You onboard it.



Where the data lives

Which systems, which fields, what's the source of truth.



What "good" looks like

The reply format, the structure, the tone of a real answer.



Your judgment criteria

How you weigh risk — the calls only experience makes.

...and you **write it all down**. The teaching *is* the work — and the work is infrastructure.

Onboard the intern — and each thing it needs is infrastructure

Same as a new hire: give it a brain, skills, hands, memory, and your knowledge — **each one is a piece of infrastructure.**

THE AGENT NEEDS...

 **A brain** — to reason & decide

 **Skills** — the tasks it can do

 **Hands** — to act in your systems

 **Memory** — to remember past work

 **Knowledge** — your tax law & data

 **Relationships** — how entities, jurisdictions & products connect

 **Context** — the task in front of it

 **Guardrails** — judgment & sign-off

...WHICH IS INFRASTRUCTURE

→ **Foundational AI model**

→ **Skill library + management**

→ **Tool calling (MCP)**

→ **Memory store**

→ **Data foundation**

→ **Knowledge graph**

→ **Grounding / retrieval**

→ **Trust & controls**

The model is just the brain. **Everything else is the infrastructure** — and that's the work.

Why the model alone isn't enough

“Did we charge the right sales tax on this \$2M order shipped to Texas?”

The model alone

Knows Texas sales tax *in general*.
But not **your** order, your nexus,
today's rate — or how to prove it.

With the foundation, it can...

Data platform	pull the real order — \$2M, ship-to TX, the product SKU
Graph	which legal entity sold it · TX state→county→city · is the product taxable
Knowledge	today's Texas rule & rate — with a citation
Models	reason over all of it to an answer
Trust & controls	check who may see it · log every step
Eval + lineage	confidence check; unsure → human · every number traces back

The model is one slice of the answer. The **infrastructure is the rest** —
and the part that holds up in an audit.

Genuinely hard — we just had a head start

Why it's hard

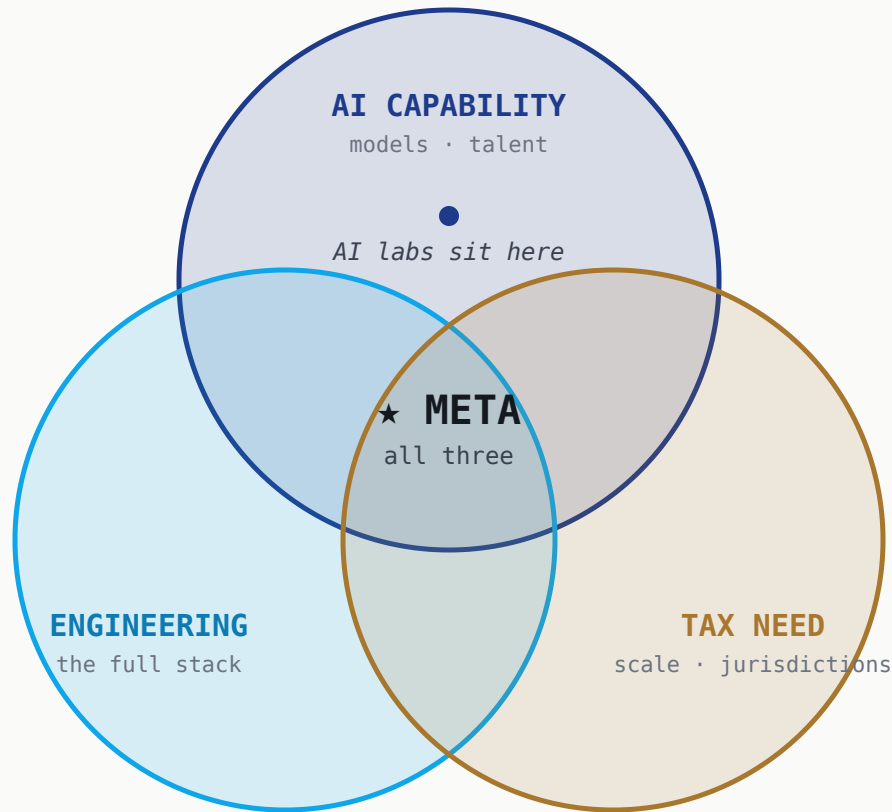
- Fragmented, messy data across ERPs & jurisdictions
- Real-time / streaming is a specialized skill set
- Talent fluent in tax *and* AI is doubly rare
- Split ownership: Tax vs. Finance Systems vs. IT
- Tax is a cost center — hard to fund infra

Why we had a head start

- Hyperscale data infrastructure already exists
- Real-time / event-driven by default
- World-class platform, data & ML engineers
- AI platform & compute in-house, not procured
- Engineering-owned tax tech — ships like a product team

The barriers most teams hit were largely **already solved here** — so we got a running start. That's luck, not genius.

The transformation takes all three at once



The frontier labs? Not yet.

- They're building the **ecosystem** — models, skills, MCP — and want you on their platform.
- But they have **many verticals to chase** — tax is just one of them.
- So they'll ship **generic skills** — but won't go deep into *your* process and data.

The horizontal AI is theirs to sell. The **vertical — your process, your data — is yours to own.** We just happened to already have all three.

THE TAKEAWAY

Tax is where coding was a few years ago.

We don't have to predict the future — we get to read it. Build the foundation now, treat the AI like the intern it is, and let your people move up to judgment.

Foundation first

Teach the intern

People move up

I'm still figuring this out. Where do you think I'm wrong?

Thank you · J Gu · PM, Tax & Finance @ Meta